

	NANYANG JUNIOR COLLEGE
	2021 JC2 Preliminary Examination H2 ECONOMICS

Paper No: 9757/01

1st September 2021
Wednesday

Time : 1400 – 1615 hrs
Duration : 2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES

Do not turn over this paper until you are told to do so.

Write your name, class and the name of your Economics tutor in the space provided on the answer booklet.

You are required to answer ALL questions.

The number of marks is given in the brackets at the end of each part question.

Write your answers on the answer booklet provided.

If you use more than one answer booklet, slot the additional booklets into the first booklet.

You are advised to spend several minutes reading through the questions and data before you begin writing your answers.

There are **7** printed pages including this cover page.

Question 1: The on-going dilemma with the miracle ingredient – Palm oil

Extract 1: How the world got hooked on palm oil

Palm oil's world domination is the result of it being able to replace less healthy fats in foods in the west, a cheap replacement of more expensive oils in home and personal care products and growing affluence in Asian countries where people consumed more fat, much of it in the form of palm oil.

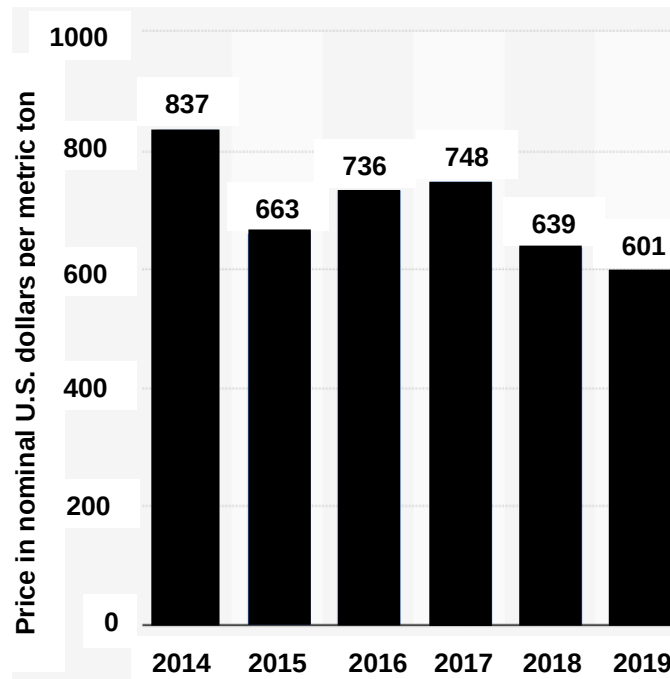
The worldwide production of palm oil is high as palm trees are perennial and evergreen which enables them to be grown year-round. In addition, the combination of favourable weather and increased fertilizer use have further contributed to the production of palm oil. Between 1995 and 2015, annual production quadrupled, from 15.2m tonnes to 62.6m tonnes. By 2050, it is expected to quadruple again, reaching 240m tonnes.

Though palm oil has many benefits, it has brought about several environmental issues which has put governments and businesses under pressure to find alternatives to palm oil. However, replacing a miracle product is not easy for example, food giant; General Mills has encountered trouble replacing palm oil in its products given its very unique physical performance characteristics that are highly challenging to duplicate.

It's clear that for now, palm oil isn't going anywhere. It's nearly impossible to avoid, and just as difficult to replace.

Source: Adapted from The Guardian, 19 Feb 2019 and BBC, 14 Jan 2020

**Figure 1: Average prices for palm oil worldwide from 2014 to 2019
(in nominal U.S. dollars per mt)**



Source: World Bank, accessed 30 May 2021

Extract 2: Tackling Indonesia's poverty with palm oil

Indonesia, the world's fourth most populous nation, and 10th largest economy in terms of purchasing power parity, has made enormous gains in poverty reduction – cutting the poverty rate by more than half since 1999, to 9.78 percent in 2020.

The palm oil industry has helped lift millions of people out of poverty, both in Indonesia and Malaysia, which together account for around 85 percent of global production. Oil palm plantations have created millions of well-paying jobs and enabled tens of thousands of smallholder farmers to own their own land. In 2019, it was estimated that the palm oil industry had succeeded in lifting 2.6 million rural Indonesians out of poverty. While at the regional level, contracts between smallholder palm oil farmers and private or state-owned companies significantly contributed to the regional economy especially at the village level in the form of infrastructure built. This not only benefited contract farmers but also non-contract farmers as well.

However, the lack of enforcement capacity by the labour authorities in Indonesia and the remote location of many plantations have contributed to child labour in the palm oil sector.

Source: Adapted from The ASEAN Post, 16 February 2021

Extract 3: The global demand for palm oil is driving the fires in Indonesia

The forests are burned deliberately by palm oil producers each year to create palm oil plantations, but this year's burns are especially destructive due to drier conditions that are causing the fires to burn out of control. Between 2001 and 2018, Indonesia lost 16% of its tree cover, or nearly 26 million hectares of forest. The loss of those forests released the equivalent of about 10.5 gigatons of carbon dioxide emissions.

A recent government audit of plantations found that more than 80% of palm oil plantations in the country are failing to comply with regulations. The audit began after smoke from the annual fires in 2015 was so bad that it sickened hundreds of thousands of people. In addition, the smoke reduced visibility to the extent that airplanes were unable to land hence causing the closure of airports in Indonesia.

Source: Adapted from Quartz, 18 September 2019

Extract 4: Who gets to define sustainable palm oil?

In 2020, Indonesia produced 59%, of the world's palm oil.

Palm oil is an important source of economic growth and livelihood in Indonesia. But it has also been associated with environmental degradation and human rights abuses. As a response to growing consumer awareness of these issues in the US and Europe, in 2004, the World Wildlife Fund (WWF), Malaysian Palm Oil Association and a few multinational companies including Unilever started the Roundtable on Sustainable Palm Oil (RSPO), a voluntary initiative that promotes sustainable production of the commodity through certification schemes and standards.

Indonesia initiated its own national certification scheme, the Indonesian Sustainable Palm Oil (ISPO) certification in 2011. In 2014, Indonesia made ISPO certification mandatory for all producers other than smallholder farmer. The final aim was to make certification mandatory for smallholders.

Palm oil producers who obtained the ISPO certification by complying with requirements such as legality, environmental accountability, social responsibility and business practices etc... are most

likely to benefit from increased sales and revenue as many large global firms such as PepsiCo, Nestle, Unilever etc... have committed to procuring only sustainable palm oil. ISPO certification gives such buyers assurance that:

1. The palm oil are produced legally
2. Palm oil producers have the requisite permits
3. Palm oil producers have followed certain environmental, labour and social standards

By the end of 2020, the Indonesian Palm Oil Association issued 682 certificates covering a total of 3.78 million hectares of land or 27% of Indonesia's planted palm oil area.

The biggest challenge to "full national coverage" comes from smallholder farmers who manage roughly 40% of Indonesia's oil palm-planted areas. Their per hectare yield can be 50% lower than that of large-scale commercial farms because of lack of access to high-quality agricultural inputs such as fertilisers, while incomplete and inefficient harvesting also play a role. Their response to the yield gap is to cut down forest to plant more palm trees. While rolling out certification schemes, helping smallholders to boost their per hectare yield on existing land is a crucial step toward sustainability.

Low education and income are often barriers for smallholder farmers to certify their palm oil. Land ownership is another hurdle. ISPO requires proper record of land ownership as a prerequisite for certification, a condition that many multi-generational smallholder farmers often cannot meet. As of October 2020, only 0.21% of oil palm planted areas controlled by smallholders were ISPO-certified.

"Indonesia is too big. Its palm oil is produced in so many different areas, which makes certification more difficult," said Robert Hii who runs CSPO Watch, an independent industry monitoring website. He added, "As Indonesia consumes 30-35% of its own palm oil, so the motivation to get certification is low. What Indonesia says is we have enough certified palm oil to satisfy the European and US market, why do we need to certify the rest?"

Source: Adapted from China Dialogue, 4 June 2021

Questions

- (a) (i) From Figure 1, identify the trend in average prices for palm oil worldwide from 2014 to 2019. [1]
- (ii) Using a diagram and Extract 1, explain the trend in average prices for palm oil worldwide identified in (a)(i). [4]
- (b) Using elasticity concept, explain the expected impact of the trend in price of palm oil identified in (a)(i) on the total revenue of palm oil producers. [3]
- (c) (i) Explain why there is a need for government intervention in the market for palm oil. [4]
- (ii) Discuss whether ISPO certification is the most effective strategy in achieving allocative efficiency in the Indonesian market for palm oil. [8]
- (d) To what extent would the changes in palm oil industry likely to lead to changes in the standard of living of people in Indonesia. [10]

[Total: 30]

Question 2: The Quest for Inclusive Growth

Extract 5: How does income inequality affect our lives?

Inequality can take different forms. If there are large numbers of poor people, economic growth may be affected by their inability to invest in education and their lower health levels, among other factors. If inequality “squeezes” the middle class, it may reduce its demand for goods and services. If inequality means rising incomes among the rich, it could see them accumulate savings, which banks can then lend out, so increasing investment in the economy. The rich may also use their economic power to lobby against policies that don’t serve their needs, for example investment in public health and education. Large wealth gaps can be associated with social conflicts, and with higher security costs, for both businesses and governments.

Source: OECD Insights, 2015

Extract 6: Rising inequality in the G7 countries

G7 countries have been facing a prolonged period of low growth and persistent slower income growth of the poorest. The poorest 20% of the G7 population receives, on average, only 5% of all income earned from work, while the richest 20 % receives about 45%. The G7 (Group of Seven) is an inter-governmental organisation made up of the world's seven largest developed economies. They are Canada, France, Germany, Italy, Japan, the UK and the United States.

Disparities exist not only in the levels of welfare, but also in how welfare has evolved over time. Multidimensional inequalities often tend to accumulate and self-perpetuate from one generation to the next, making it harder for disadvantaged children to climb the socio-economic ladder and perpetuating a vicious cycle. On average, in the G7, children with lower-educated parents have just a 15% chance of attaining tertiary education, set against 63% for children whose at least one parent had attained tertiary education. As a result, social mobility is low in many of the G7 economies.

A wide range of factors like skill-biased technological change, economic globalisation and structural changes in the labour market have contributed to rising and/or persistently high inequalities. In addition, the tax and benefit systems became less redistributive between the mid-1990s and mid-2000s exacerbating income inequality. The main reason was tighter unemployment benefit eligibility rules which reduced the number of people entitled to transfers.

Source: Adapted from OECD.org, 2017

Extract 7: The G7's Deadly Sins: Burning the planet

The climate crisis is inextricably linked to economic inequality: it is driven by the greenhouse gas emissions caused by the unsustainable development model that for more than a century has enriched the ‘haves’ at the expense of the ‘have-nots’ and the planet. G7 countries have a huge responsibility in facilitating the transition to climate-friendly development pathways. Historically, they have been responsible for the largest amount of emissions and their economies are still dependent on ever-growing consumption.

With the exception of Italy, six G7 countries are among the top 10 countries most responsible for cumulative carbon emissions. Approximately half of total global emissions from consumption can be attributed to the richest 10% of people. Of these emissions, 77% are generated by the richest 10% of people living in G7 countries. G7 countries are also responsible for failing to take action to cut emissions. Even Germany, often wrongly perceived as a climate leader, will miss its target to cut emissions by 40% by 2020, largely because of inaction in the transport and construction sectors.

Source: Adapted from Oxfamilibrary, 2019

Extract 8: G7 Unemployment facts and figures

The unemployment rate in G7 countries increased to 6.7% in 2020 compared to 4.3% in 2019. The youth unemployment rate (15-24), at 13.9% in 2020, remains a major concern. The gender gap in the labour force participation rate stood at 12.8 percentage points in 2020, down from 14.1 percentage points in 2014.

The share of youth not in employment, education or training (NEET) was estimated at 10.6% for male and at 11.2% for female in 2019 down from 12% for male and 13.3% for female in 2015. An estimated 7.8% of the adult (25+) and 18.1% of the youth (15-24) extended labour force was underutilized in 2020.

Source: ILOSTAT, ILO modelled estimates, April 2021

Table 1: Debt of G7 Nations

G7 countries	General Government Net Debt/GDP (%) (2019)
Canada	23
Germany	41
Uk	75
Europe	89
USA	83
Italy	122
Japan	150

Source: icaew.com/insights April 2021

Table 2: Selected economic indicators of Singapore

	2015	2016	2017	2018	2019	2020
Real GDP growth rate (%)	2.98	3.33	4.52	3.5	1.35	-5.39
Inflation rate (%)	-0.52	-0.53	0.58	0.44	0.57	-0.18
Unemployment rate (%)	3.79	4.08	4.2	3.65	3.1	5.19
Budget position as a percentage of GDP (%)	1.7	3.66	5.31	3.69	3.84	-8.94
Gini coefficient (before accounting for government transfers & taxes)	0.464	0.458	0.459	0.458	0.452	0.452
Gini coefficient (after accounting for government transfers & taxes)	0.409	0.401	0.402	0.403	0.398	0.375

Source: Statista.com & Singstat.gov.sg

Extract 9: Why Singapore gives top priority to fighting income inequality?

Singapore's politics will turn vicious, its society will fracture and the country will wither if it allows widening income inequalities to create "a rigid and stratified social system", said Prime Minister Lee Hsien Loong in a reply to a parliamentary question from Mr Gan Thiam Poh.

"As globalisation and technological disruption have widened income inequality, the Government has over the years intervened more aggressively to support the less well-off," he said, citing both long-term policies such as education, home ownership and affordable healthcare, as well as

targeted, means-tested programmes which provide transfers and subsidies to lower-income groups such as the Workfare Income Supplement scheme.

Education is a critical plank of the Government's efforts, he stressed, whether in building up pre-schools and having schemes like KidStart for children from poor families or giving out bursaries and getting people to go for training via SkillsFuture. However, he also cautions that it will get harder to narrow and bridge class divisions as our society matures.

He stressed that the government policies on urban planning and public housing have enabled ethnic and social integration, and distributed access to good schools, healthcare, parks and recreation across the island.

To fund increased social spending, "significant changes" have been made, like introducing GST in 1994 as a source of revenue, he said. "But though we have far more extensive social safety nets now than we did in the 1970s and 1980s, it is important to strike the right balance: Providing sufficient transfers to support those who need extra help so they can help themselves, but without diminishing their incentive to work or discouraging enterprise."

Source: The Straits Times, 6 February 2018

Questions

- (a) With reference to Table 2, compare the change in Gini coefficient (before accounting for government transfers and taxes) with that of the change in the Gini coefficient (after accounting for government transfers and taxes) for Singapore from 2015 to 2020. [2]
- (b) Explain how "skill-biased technological change" leads to wage differential as referred to in Extract 6. [2]
- (c) With the help of a diagram, explain why there exists misallocation of resources in the market for tertiary education. [4]
- (d) (i) Explain how a less distributive tax and benefit system will impact the real national income of an economy. [4]
(ii) Discuss whether attaining low rate of unemployment should be the main aim for G7 countries. [8]
- (e) Discuss whether supply-side policies are more appropriate than demand-management measures in achieving inclusive growth in Singapore. [10]

[Total: 30]